

CDK12 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes CDK12 gene fusions from the msk_impact_50k_2026 study, covering 43 fusion events. 7/43 fusions (16.3%) were in-frame. 16/43 fusions (37.2%) retained the PF00069 domain. Domain retention was tested with Fisher's exact test ($p=0.0677006$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for CDK12.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 43 analyzed fusion events, identifying 29 distinct fusion partner genes. The most frequent partner was FBXL20, observed in 6 events.

1.2 Domain retention

PF00069 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 5/7 (71.4%) of in-frame fusions retained the domain, $p=0.0677006$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.316832.

1.3 Domain disruption

Domain-disruption analysis was skipped: CDK12 has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 41 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 644 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -84 aa from the nearest configured domain boundary at 728 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: CDK12 has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: CDK12 has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
ADCY9	2	0.046511627906976744	0.04991713954627366		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.25204681894094494	1
AKAP8L	1	0.023255813953488372	0.04991713954627366		0.9906291834002677		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.24007200940379297	2
HTR1E	2	0.046511627906976744	0.04991713954627366		0.9250334672021419		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23598256191283248	3
MED13	1	0.023255813953488372	0.04991713954627366		0.9451137884872824		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.23031871049386757	4
ODAD4	1	0.023255813953488372	0.04991713954627366		0.9210174029451138		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22515519930626002	5
TBKBP1	1	0.023255813953488372	0.04991713954627366		0.9210174029451138		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.22515519930626002	6
GRB7	1	0.023255813953488372	0.04991713954627366		0.85809906291834		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2116726978719513	7
ERBB2	3	0.06976744186046512	0.04991713954627366		0.7367246764837126		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.20559745559609263	8
RECQL4	1	0.023255813953488372	0.04991713954627366		0.7817938420348058		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.19532157911119402	9
STAT5B	4	0.09302325581395349	0.04991713954627366		0.5917001338688086		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1844875453015368	10
COLEC12	1	0.023255813953488372	0.04991713954627366		0.6974564926372155		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.17724928995456754	11
CACNA1G	1	0.023255813953488372	0.04991713954627366		0.6613119143239625		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16950402317315616	12
FBXL20	6	0.13953488372093023	0.04991713954627366		0.4152164212405176		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16660316169846445	13
SYNRG	1	0.023255813953488372	0.04991713954627366		0.6265060240963856		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.16204561812438967	14
ANKFN1	1	0.023255813953488372	0.04991713954627366		0.6050870147255689		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.1574558304020718	15
SKAP1	1	0.023255813953488372	0.04991713954627366		0.5917001338688086		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.15458721307562318	16
USP6	1	0.023255813953488372	0.04991713954627366		0.5917001338688086		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.15458721307562318	17
CSF3	1	0.023255813953488372	0.04991713954627366		0.5394912985274432		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.14339960550247344	18
FAM222B	1	0.023255813953488372	0.04991713954627366		0.4029451137884873		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.11413970877269718	19
IKZF3	2	0.046511627906976744	0.04991713954627366		0.321285140562249		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.10660792048999827	20
SPON1	1	0.023255813953488372	0.04991713954627366		0.32797858099062915		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.09807545174458471	21
SEC31A	1	0.023255813953488372	0.04991713954627366		0.2449799196787149		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08029002432060309	22
VMP1	1	0.023255813953488372	0.04991713954627366		0.2449799196787149		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.08029002432060309	23
SNF8	1	0.023255813953488372	0.04991713954627366		0.18340026773761708		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.06709438461893927	24
MAP3K3	2	0.046511627906976744	0.04991713954627366				['domain_retention', 'recurrence']	0.04805958774302079	25

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
MTRES1	1	0.023255813953488372	0.04991713954627366		0.048192771084337394		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.03812134962180791	26
IGFBP4	1	0.023255813953488372	0.04991713954627366		0.01740294511378848		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.031523529770976	27
FGFR4	1	0.023255813953488372	0.04991713954627366		0.01338688085676043		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.030662944573041417	28
RPS6KB1	1	0.023255813953488372	0.04991713954627366		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.027794327246592757	29

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
80	1	15	0	25		0.39024390243902435	0.4086638740638108
240	2	14	0	25		0.14634146341463414	0.8346326063360919
246	3	13	0	25		0.052532833020637895	1.279569177684353
349	4	12	0	25		0.01797175866495507	1.7454094219943264
603	5	11	0	25		0.0058286784859313745	2.2344299000136965
637	6	10	0	25		0.0017809850929234755	2.749339715622759
644	9	7	0	25		3.265366098560993e-05	4.486068120461908
700	9	7	2	23	14.785714285714286	0.0011510086327457892	2.938921419081068
703	11	5	2	23	25.3	7.698377534159563e-05	4.113600794594795
750	11	5	3	22	16.133333333333333	0.0004274646628148892	3.369099781237133
807	11	5	4	21	11.55	0.0009913088581971262	3.003791013009668
870	11	5	5	20	8.8	0.003014843556286022	2.520735218981372
897	11	5	6	19	6.966666666666667	0.008600185164931898	2.0654921981468433
923	11	5	9	16	3.911111111111111	0.05784571299099636	1.2377288215582707
949	11	5	15	10	1.4666666666666666	0.7420848641039097	0.12954642638599925
955	11	5	16	9	1.2375	1.0	-0.0
988	11	5	17	8	1.035294117647059	1.0	-0.0
1032	12	4	17	8	1.411764705882353	0.7341702326198091	0.13420322820990868
1090	12	4	18	7	1.1666666666666667	1.0	-0.0
1121	12	4	19	6	0.9473684210526315	1.0	-0.0
1146	13	3	19	6	1.368421052631579	1.0	-0.0
1159	13	3	20	5	1.0833333333333333	1.0	-0.0
1208	13	3	22	3	0.5909090909090909	0.6623956829348356	0.17888250666352729
1254	15	1	22	3	2.0454545454545454	1.0	-0.0
1355	16	0	22	3		0.26829268292682923	0.5713911715615105
1378	16	0	23	2		0.5121951219512195	0.2905645619858162
1381	16	0	24	1		1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

5	11
2	23

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein kinase domain	41	16	15	10	0.22484641638225256

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
ADCY9	2
AKAP8L	1
ANKFN1	1
CACNA1G	1
COLEC12	1
CSF3	1
ERBB2	3
FAM222B	1
FBXL20	6
FGFR4	1
GRB7	1
HTR1E	2
IGFBP4	1
IKZF3	2
MAP3K3	2
MED13	1
MTRES1	1
ODAD4	1
RECQL4	1
RPS6KB1	1
SEC31A	1
SKAP1	1
SNF8	1
SPON1	1
STAT5B	4
SYNRG	1
TBKBP1	1
USP6	1
VMP1	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	prob_intron	prob_intron_2	prob_intron_3	main_retained_frac	non_retained_frac	domain_disrupted	source_annotation	effect_on_fusion
EVT-P-0039125-T01-IM6-1	P-0039125-T01-IM6		AKAP8L-CDK12	AKAP8L	unknown	three_primes	37627994	2	637	False	retained	1.0	False	Protein kinase domain		AKAP8L (NM_014371) - CDK12 (NM_016507) rearrangement: t(17;19)(q12;p13.12)(chr17:g.37627994::chr19:g.104629::CDK12) Protein Fusion: {AKAP8L:CDK12}	NA
EVT-P-0056622-T01-IM6-2	P-0056622-T01-IM6		ANKFN1-CDK12	ANKFN1	in-frame	three_primes	37619399	1	349	True	retained	1.0	False	Protein kinase domain		ANKFN1 (NM_153228) - CDK12 (NM_016507) rearrangement: c.62+8631:ANKFN1_c.1046+29:CDK12 Protein Fusion: in frame {ANKFN1:CDK12}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	breakpoint_4	main_ref	donor_frame	maintained_domains	disrupted_domains	description	annotation	effect_on_fra	
EVT-P-0003762-T02-IM5-4	P-0003762-T02-IM5		CDK12-ADCY9	ADCY9	unknown	three_prime	37646665	3	644	True	retained	1.0	False	Protein kinase domain			ADCY9 (NM_001116) - CDK12 (NM_016507) rearrangement: t(16;17)(p13.3;q12)(chr16:g.4033430::chr17:g.1569966) Protein fusion: mid-exon (ADCY9-CDK12)	NA	
EVT-P-0003762-T01-IM5-5	P-0003762-T01-IM5		CDK12-ADCY9	ADCY9	unknown	three_prime	37646665	3	644	True	retained	1.0	False	Protein kinase domain			t(16;17)(p13.3;q12)(chr16:g.4033430::chr17:g.1569966) Protein fusion: mid-exon (ADCY9-CDK12)	NA	
EVT-P-0024871-T01-IM6-7	P-0024871-T01-IM6		CDK12-CACNA1G	CACNA1G	unknown	five_prime	37667806	8	897	False	disrupted	0.580204778	0.569966		Protein kinase domain		CDK12 (NM_016507) - CACNA1G (NM_018896) rearrangement: c.2691:CDK12_c.6399+226:CACNA1Gd	NA	
EVT-P-0038990-T01-IM6-53	P-0038990-T01-IM6		CDK12-FBXL20	FBXL20	in-frame	five_prime	37676367	11	1032	True	retained	1.0	False	Protein kinase domain			Fusion: mid-exon {CDK12:CACNA1G} CDK12 (NM_016507) - FBXL20 (NM_032875) rearrangement: c.3095+27:CDK12_c.104+18457:FBXL20 Protein Fusion: in frame {CDK12:FBXL20}	NA	
EVT-P-0002174-T01-IM3-55	P-0002174-T01-IM3		CDK12-GRB7	GRB7	unknown	three_prime	37650768	5	750	True	disrupted	0.924914675	0.779181		Protein kinase domain		NA NA Transcript fusion (GRB7-CDK12)	NA	
EVT-P-0008960-T02-IM6-56	P-0008960-T02-IM6		CDK12-HTR1E	HTR1E	unknown	five_prime	37646977	3	700	False	lost	0.0	False	Protein kinase domain			CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697113::chr17:g.1569966) Protein Fusion: mid-exon {CDK12:HTR1E}	NA	
EVT-P-0008960-T01-IM5-57	P-0008960-T01-IM5		CDK12-HTR1E	HTR1E	unknown	five_prime	37646977	3	700	False	lost	0.0	False	Protein kinase domain			CDK12 (NM_016507) - HTR1E (NM_000865) rearrangement: t(6;17)(q14.3;q12)(chr6:g.87697112::chr17:g.1569966) Protein fusion: mid-exon (CDK12-HTR1E)	NA	
EVT-P-0060111-T01-IM7-58	P-0060111-T01-IM7		CDK12-IKZF3	IKZF3	in-frame	five_prime	37682742	13	1254	True	retained	1.0	False	Protein kinase domain			CDK12 (NM_016507) - IKZF3 (NM_012481) rearrangement: c.3760+173:CDK12_c.8-12056:IKZF3inv Protein Fusion: in frame {CDK12:IKZF3}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	chr	start	end	domain	disrupted	description	annotation	effect_on_fra
EVT-P-0016507-T03-IM6-75	P-0016507-T03-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	9	949	True	disrupted	0.245733788	2560443				Protein kinase domain		CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel.	NA	
EVT-P-0016507-T02-IM6-76	P-0016507-T02-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	9	949	True	disrupted	0.245733788	2560443				Protein kinase domain		CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel.	NA	
EVT-P-0016507-T01-IM6-77	P-0016507-T01-IM6		CDK12-STAT5B	STAT5B	unknown	three_prime	37672113	9	949	True	disrupted	0.245733788	2560443				Protein kinase domain		CDK12 (NM_016507) - STAT5B (NM_012448) rearrangement: c.2846+52:CDK12_c.2238-225:STAT5Bdel.	NA	
EVT-P-0023516-T01-IM6-78	P-0023516-T01-IM6		COLEC12-CDK12	COLEC12	out-of-frame	three_prime	37665943	7	870	True	disrupted	0.515358361	774744				Protein kinase domain		COLEC12 (NM_130386) - CDK12(NM_016507) rearrangement: t(17;18)(q12;p11.32)(chr17:g.37665943::	NA	
EVT-P-0038406-T01-IM6-80	P-0038406-T01-IM6		CSF3-CDK12	CSF3	out-of-frame	three_prime	37676083	11	988	True	disrupted	0.112627986	31612286				Protein kinase domain		{COLEC12} CSF3 (NM_000759) - CDK12 (NM_016507) Rearrangement: c.313-58:CSF3_c.2964-126:CDK12dup	NA	
EVT-P-0046790-T01-IM6-81	P-0046790-T01-IM6		ERBB2-CDK12	ERBB2	out-of-frame	three_prime	37640037	3	644	True	retained	1.0	False				Protein kinase domain		{CSF3:CDK12} ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.225+207:ERBB2_c.1932-6773:CDK12dup	NA	
EVT-P-0043706-T01-IM6-82	P-0043706-T01-IM6		ERBB2-CDK12	ERBB2	unknown	three_prime	37668120	8	923	True	disrupted	0.334470989	76109213				Protein kinase domain		{ERBB2:CDK12} ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.2293:ERBB2_c.2768+237:CDK12dup	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint	breakpoint	protein	protein	breakpoint	main_ref	chr	start	end	domain	disrupted	description	annotation	effect_on_fra
EVT-P-0037167-T01-IM6-83	P-0037167-T01-IM6		ERBB2-CDK12	ERBB2	unknown	three_prime	37673710	10	955	False	disrupted	0.225255972	0.0624573				Protein kinase domain		ERBB2 (NM_004448) - CDK12 (NM_016507) rearrangement: c.2553:ERBB2_c.2864:CDK12dup Protein Fusion: mid-exon {ERBB2:CDK12} CDK12 NA (NM_016507) rearrangement: c.3269:CDK12_chr17:g.27096697inv Protein Fusion: mid-exon {FAM222B:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.43-9338:FBXL20_c.3362:CDK12inv Protein fusion: mid-exon (FBXL20-CDK12) FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.398+4667:FBXL20_c.2769-267:CDK12inv Protein Fusion: in frame {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA	CDK12dup
EVT-P-0036357-T01-IM6-84	P-0036357-T01-IM6		FAM222B-CDK12	FAM222B	unknown	three_prime	37681100	12	1090	False	lost	0.0	False				Protein kinase domain		CDK12 NA (NM_016507) rearrangement: c.3269:CDK12_chr17:g.27096697inv Protein Fusion: mid-exon {FAM222B:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.43-9338:FBXL20_c.3362:CDK12inv Protein fusion: mid-exon (FBXL20-CDK12) FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.398+4667:FBXL20_c.2769-267:CDK12inv Protein Fusion: in frame {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA	27096697inv
EVT-P-0014403-T01-IM6-86	P-0014403-T01-IM6		FBXL20-CDK12	FBXL20	unknown	three_prime	37682171	13	1121	False	lost	0.0	False				Protein kinase domain		CDK12 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.43-9338:FBXL20_c.3362:CDK12inv Protein fusion: mid-exon (FBXL20-CDK12) FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.398+4667:FBXL20_c.2769-267:CDK12inv Protein Fusion: in frame {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA	CDK12inv
EVT-P-0046675-T01-IM6-87	P-0046675-T01-IM6		FBXL20-CDK12	FBXL20	in-frame	three_prime	37671717	9	923	True	disrupted	0.334470989	0.109213				Protein kinase domain		FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.398+4667:FBXL20_c.2769-267:CDK12inv Protein Fusion: in frame {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA	FBXL20_c.2769-267:CDK12inv
EVT-P-0043910-T01-IM6-88	P-0043910-T01-IM6		FBXL20-CDK12	FBXL20	unknown	three_prime	37682285	13	1159	False	lost	0.0	False				Protein kinase domain		FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+20635:FBXL20_c.3476:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA	CDK12inv
EVT-P-0036553-T01-IM6-89	P-0036553-T01-IM6		FBXL20-CDK12	FBXL20	unknown	three_prime	37619062	1	246	False	retained	1.0	False				Protein kinase domain		FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.697-2344:FBXL20_c.738:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12} FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA	CDK12inv
EVT-P-0048892-T01-IM6-90	P-0048892-T01-IM6		FBXL20-CDK12	FBXL20	unknown	three_prime	37682432	13	1208	False	lost	0.0	False				Protein kinase domain		FBXL20 NA (NM_032875) - CDK12 (NM_016507) rearrangement: c.42+25196:FBXL20_c.3623:CDK12inv Protein Fusion: mid-exon {FBXL20:CDK12}	NA	CDK12inv

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_coding	in_frame	main_retained	donor_inframe	unintended_domains	disrupted_domains	domains_annotation	effect_on_fra
EVT-P-0041371-T01-IM6-99	P-0041371-T01-IM6		TBKBP1-CDK12	TBKBP1	in-frame	three_prime	37648975	4	703	True	retained	1.0	False	Protein kinase domain		TBKBP1 (NM_014726) - CDK12 (NM_016507) rearrangement: c.872+1376-TBKBP1_c.2109-29:CDK12dup Protein Fusion: {TBKBP1:CDK12}	NA
EVT-P-0019776-T01-IM6-100	P-0019776-T01-IM6		USP6-CDK12	USP6	in-frame	three_prime	37673592	10	949	True	disrupted	0.245733788	0.3560443		Protein kinase domain	USP6 (NM_004505) - CDK12 (NM_016507) rearrangement: c.2916-470:USP6_c.2847-101:CDK12del Protein Fusion: in frame {USP6:CDK12}	NA
EVT-P-0059578-T01-IM7-101	P-0059578-T01-IM7		VMP1-CDK12	VMP1	unknown	three_prime	37618564	1	80	False	retained	1.0	False	Protein kinase domain		VMP1 (NM_030938) - CDK12 (NM_016507) rearrangement: c.582+1747-VMP1_c.240:CDK12dup Protein Fusion: mid-exon {VMP1:CDK12}	

3 Figures

CDK12 domain-retention track

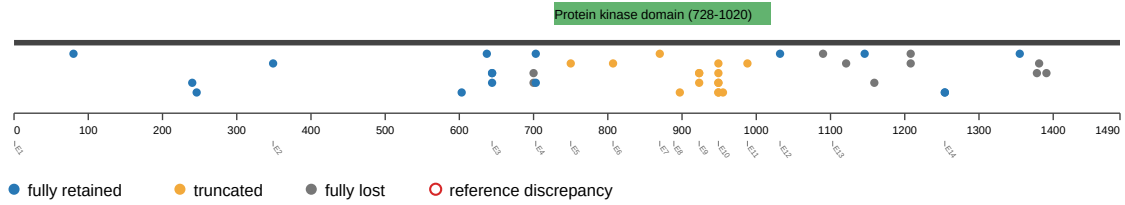
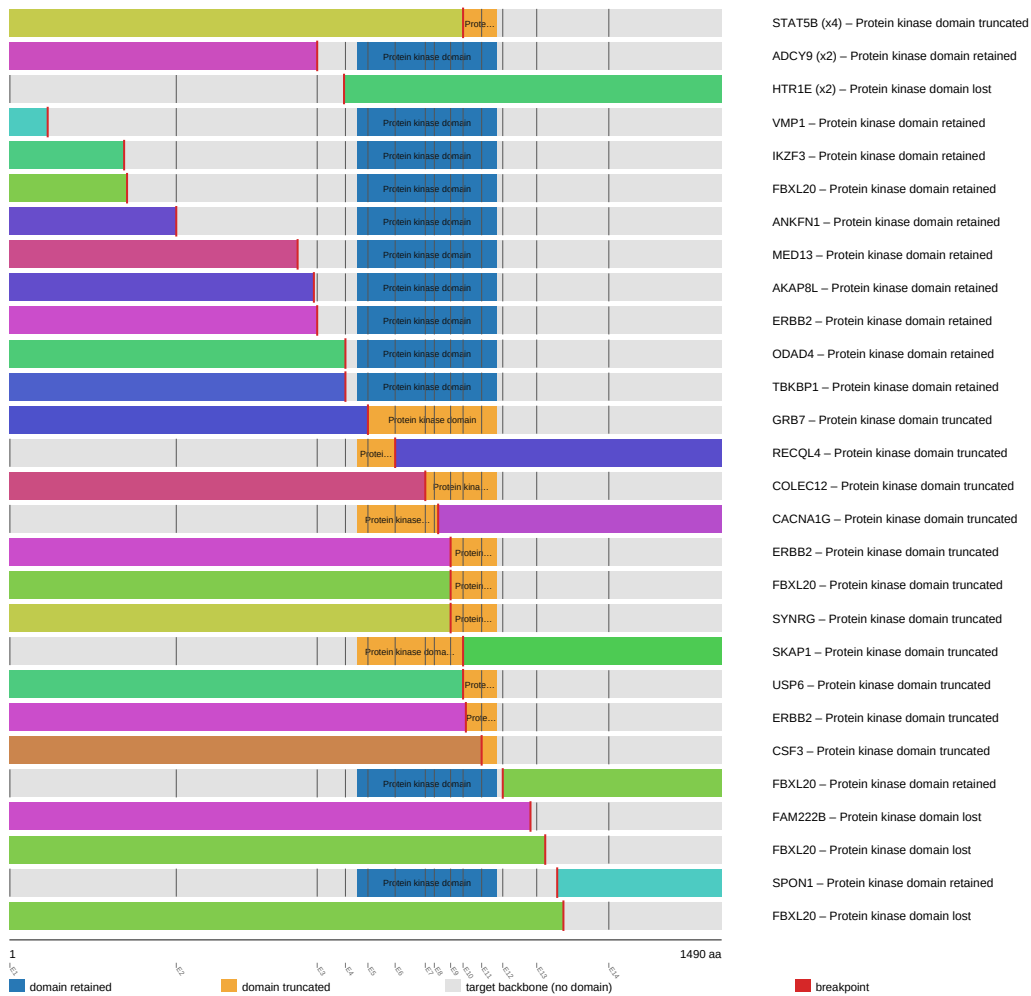


Figure 1: domain retention outliers

CDK12 fusion-transcript schematic

partner block → breakpoint → retained CDK12 portion (domain-colored, exon ticks)



Showing the top 28 of 36 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

Reference vs msk_impact_50k_2026



Figure 3: reference comparison